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Subjective and objective assessments of executive functions are independently predictive of aggressive tendencies in patients with substance use disorder

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ABSTRACT

Background and aims: Impairments in executive functions have been found to influence violent behavior. Executive functions are crucial in the treatment of patients with substance use disorders because substance use generally impairs cognitive processes and is therefore detrimental for executive functions thereby reducing control of behavior and thus of consumption impulses. We studied correlations between subjective, i.e. self-report, and objective, i.e. behavior-based, assessment of executive functions and the predictive validity of these measures for aggression in patients with substance use disorder.

Methods: The study included 64 patients with a diagnosed substance use disorder who were convicted according to the German Criminal Code for crimes they committed in the context of their disorder and were therefore in treatment in forensic psychiatric departments in Germany. Multiple self-report and behavior-based instruments were used to assess executive functions, appetitive and facilitative aggression as well as clinical and sociodemographic variables.

Results: Participants showed impaired executive functions, and measures of executive functions predicted aggressive tendencies and violent offenses. Despite ecological validity of the findings, the subjective and objective assessments of executive functions did not correlate with each other, which corroborates studies in other clinical settings.

Conclusions: We discuss that this finding may be due to the conceptual differences between subjective and objective measures. Therefore, self-report and behavior-based measures should not be used as proxies of each other but as complementary measures that are useful for comprehensive diagnostics of cognitive impairments and assessment of risks for violent behavior.

1. Introduction

Executive functions are generally impaired in patients with substance use disorder [SUD, 1], and chronic use of psychoactive substances persistently affects cognitive structures in the brain [2,3]. For example, cognitive deficits caused by alcohol are still detectable after an abstinence period of one year or more [4]. Successful recovery from SUD requires effective executive control functions, such as inhibition, planning or initiation of behavior [5], but a central determinant of SUD is loss of control. Thus, the effects of SUD on brain functions strengthen one of its basic causes. Moreover, a constellative factor of SUD is aggression, which also correlates with a specific profile of executive functions and specific neurobiological structures [see 6 for a review]. Meijers et al. [7] found significant differences in executive functions between violent and non-violent offenders. Taken together, the above findings show that impulsive behavior and weakened inhibitory functions can lead to an increased propensity towards aggressive behavior.

The importance of these findings and their interaction is therefore to be considered in treatment of patients with SUD. A broad neuropsychological assessment of patients with SUD [see 3 for a review] is considered a necessary component in diagnostics. There are two general classes of neuropsychological tests for the assessment of executive

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functions: Self-rating measures, which gather information through questionnaires that ask testees to self-rate the extent of their executive capabilities, mostly operationalized as functioning in everyday activities; and performance-based measures, which assess executive capabilities by collecting accuracy or reaction time data. Both kinds of tests provide valuable insights, but their results do not always correlate despite the apparent thematic relationship of their constructs [8–13]. Toplak et al. [12] assume that self-rating and performance-based measures tap different levels of analysis following Marr's [14] conceptualization of the analysis of information processing. This reasoning would imply that the different methods of assessment actually reflect different constructs. Nevertheless, the two methods both offer important information that can only be acquired via either the one or the other way. Rating measures, especially those relying on self-report, allow to gain insights into the individual introspection of the studied construct; of central interest here is the subjective estimation of a cognitive capacity based on individual experience. With performance-based tests cognitive capacity can be quantified objectively, so measurements are less affected by conscious thoughts, opinion, or attitudes of the individual subject; consequently, the results of such tests can be considered to be purer measures of the studied construct.

The therapeutic value of utilizing both approaches lies in the possibility to ascertain on the one hand, if an individual has any cognitive impairments at all, and on the other hand to learn, if and in what way the concerned individual is aware of these impairments. Rothlind et al. [10] discuss that the degree of awareness of a cognitive deficit is predictive of the therapeutic outcome and that the causality seems obvious: Patients who are not aware of a problem will be less inclined to invest in order to improve the deficit.

The critical question for our study was to further examine whether subjective and objective assessments of executive functions align with each other and whether they predict aggression – either in the form of a tendency or a propensity towards aggressive behavior, which can be assessed by self-report questionnaires, or in the form of actual committed acts of violence, which can be assessed by recording violent crimes such as manslaughter or assault.

We compared subjective self-assessment in the Behavior Rating Inventory of Executive Function – Adult Version [BRIEF-A, [15]] and objective performance measurements in a neuropsychological test battery for cognitive functions [CogBat, [16]] in patients with a diagnosis of SUD who had been convicted of a criminal offense causally connected to the SUD and were therefore in treatment in a forensic psychiatric hospital.

Executive Functions generally include the "processing related to goal-directed behavior, or the control of complex cognition, especially in nonroutine situations" [17]. An influential concept of an executive system was formulated as the Central Executive by Baddeley and Hitch [18] in their Working Memory theory. The Central Executive was attributed the role of controlling action [19]. It is structurally similar to the Supervisory Attentional System of Norman and Shallice [20] that more explicitly states that cognitive control is only involved in planning, organizing novel sequences of actions, or overriding strong stimulus-response pairings while most processes run automatically without executive monitoring. Experimental psychology elaborated such concepts to study the control of action; neuropsychology and clinical psychology [21], however, studied executive functions as heterogeneous processes [22]. The question concerning the unity and diversity of executive functions reaches back for over 50 years [23,24] and is still a matter of current debate [17,25]. Executive tasks are generally considered to involve frontal areas, although each specific task may recruit specific areas additionally. McCabe et al. [17] presented results that point towards a common executive component in the diverse executive functions. The concept of executive function as used in context of this study adheres to this idea. The functions studied can be subsumed under a general control system but also including subsystems with specialized functions.

A comparison between subjective and objective measures of these functions is both theoretically and practically interesting as it challenges the idea of an unitary construct of executive functions from a metacognitive perspective. Theoretical considerations thus center around the question whether conscious monitoring of these cognitive processes is entirely possible [see [26] for a review]. Toplak et al. [12] discuss whether the level of analysis of the underlying constructs is identical. The question whether or not testees can scan their entire mental experience and whether or not the construct of both types of test is identical has valuable practical implications. Self-report instruments require testees to make decisions about executive functions on the basis of the information they can consciously grasp; this consciously attainable information also forms the basis of their decisions in everyday life, so it is of high relevance. The value of comparing subjective and objective measures is that such a comparison enables one to adopt a metacognitive perspective. The question whether patients with cognitive impairments can correctly gauge their capabilities is relevant as it determines the extent of motivational involvement of patients in therapy. For example, Sadeghi et al. [27] describe deficits in both cognitive functions and metacognitive judgment in patients with an addiction [see also [28], for a review]. Meijers et al. [7] found cognitive impairments in violent offenders, but the causal connection is unclear. Nevertheless, it is of high therapeutic value to examine possible correlations between SUD, deficits in executive functions, and tendencies towards aggressive behavior.

According to Weierstall and Elbert aggression can consist of reactive or appetitive aggression; the former can be described as a "response to a certain threat and in defense of oneself, property, or other people," and the latter, "as the perpetration of violence and/or the infliction of harm to a victim for the purpose of experiencing violence-related enjoyment" [29]. Assessing the risk of aggression in a patient group with SUD is essential when evaluating therapeutic progress in forensic-psychiatric treatment. The aim of the present study was to analyze the extent of cognitive impairment in forensic patients, their subjective assessment of these impairments, and the relations of these measures to their propensity towards aggression and actual violent crimes. Specifically, we hypothesized, that the clinical sample would show impairments of executive functions in subjective as well as objective assessments of these functions; that subjective and objective assessments of executive functions would correlate with each other; and finally, that subjective and objective assessments of executive functions would predict propensity towards aggression and actual acts of violence.

2. Methods

2.1. Participants

Sixty-four male patients from the forensic psychiatric departments of the district hospitals Günzburg ($n = 44$) and Kaufbeuren ($n = 20$), Bavaria, Germany, participated in this study. Participants' age ranged from 20 to 62 years ($M \pm SD$, 34 ± 9.70 years). All participants were able to speak German sufficiently well. They were recruited for the present and another, previously reported study [30]. The participants of this study were convicted for crimes in the context of a SUD according to section 64 of the German Criminal Code. The German legal term "Hang", translated in the following with "tendency", comprises substance-related diagnoses that correspond to the concept of SUD of the DSM-V [31]. Under German criminal law, admission to a forensic psychiatric hospital is based on a court decision. If a person has a tendency to consume intoxicating substances to excess and is convicted of an unlawful act that they committed while intoxicated or that was attributable to their tendency to consume intoxicating substances, the court orders placement in a forensic psychiatric hospital if there is a risk that as a result of their tendency, the person will commit significant unlawful acts again. The only exclusion criterion was a clinically manifested thought disorder, although no potential participants fulfilled this criterion. The study was performed in accordance with the Declaration of

Helsinki and approved by the ethics review board of the University of Ulm (No. 497/18; February 6, 2019). All participants gave written informed consent. They were informed that they could terminate the study without having to give a reason. Participation was not remunerated.

2.2. Data acquisition

Data were collected by scientifically trained personnel, who visited the forensic psychiatric departments of the two participating hospitals. Patients received information about the study aims and procedures, and any of their questions or concerns were addressed. Local examination and conference rooms were used for data collection. Each session lasted about 30 to 50 min. Before data acquisition, each participant had to sign three different declarations of consent: consent to participate, the German privacy protection act, and consent to sampling of body fluids, the results of the latter were published in another study [cf. 30].

2.3. Questionnaires

2.3.1. Sociodemographic, clinical, and forensic characteristics

Participants were asked to provide information on their age, level of education, current diagnosis, index criminal offenses, and length of the current treatment in months.

2.3.2. Assessment of appetitive and reactive aggression

We assessed the tendency to violent behavior as the scores in the two AFAS subtests *reactive aggression* and *appetitive aggression* with the German version of the Appetitive and Facilitative Aggression Scale [AFAS, [29]]. The AFAS is a modified version of the Appetitive Aggression Scale, which uses 15 items to assess appetitive aggression in people who have experienced extreme violence in their lives. Weierstall and Elbert [29] report significant correlations of the Appetitive Aggression Scale with the Aggression Questionnaire [32] thus assuming construct validity. The 15 questions on appetitive aggression were adapted by Weierstall and Elbert to a civilian setting and supplemented by 15 additional items that ask about reactive aggression styles. The testees rate how often they have behaved aggressively or had aggressive feelings or thoughts. They responded to the items on a 5-point Likert scale ranging from 0 (never) to 4 (very often). We summed the items to calculate separate scores for appetitive aggression and reactive aggression, each ranging from 0 to 60, whereby a higher score indicated a higher level of aggression. An example of a question about appetitive aggression would be, “Have you been so fascinated by a fight that you couldn’t stop fighting?” and for reactive aggression “Have you attacked someone physically because you were provoked?”. In the present study, internal consistency for the Appetitive Aggression subscale was α = .90 and for the Reactive Aggression Scale α = .94.

2.3.3. Self-rating of executive functions

The BRIEF-A [15] was used for the self-report assessment of executive function. The BRIEF-A comprises nine basic scales that are summed to calculate the Global Executive Composite (GEC) score. The basic scales can also be summarized in two indexes, the Behavioral Regulation Index (BRI) and the Metacognition Index (MI). The BRI includes the Inhibit, Shift, Emotional Control, and Self-Monitor scales, and the MI comprises the Initiate, Working Memory, Plan/Organize, Task Monitor, and Organization of Materials scales. Table 1 provides a summary of the constructs operationalized with these scales. The self-report BRIEF-A has moderate to high internal consistency (alpha range, .93 to .96 for indexes and GEC). Test-retest correlations range from .82 to .93 over an average interval of 4.22 weeks. In terms of convergent validity, the self-report BRIEF-A scales, GEC score, and BRI and MI indexes demonstrate significant correlations in the expected direction with self-reports on the Frontal Systems Behavior Scale, Dysexecutive Questionnaire, and Cognitive Failures Questionnaire. Factor analysis yielded a 2-factor solution (i.e., Behavioral Regulation, Metacognition) for normative and mixed clinical/healthy adult samples, accounting for 73% and 76% of the variance, respectively [15]. In the present study the superordinate measures of the BRIEF had the following internal consistencies: GEC α = .94, BRI α = .88 and MI α = .92; the internal consistencies of the subscales were: Inhibit α = .64, Shift α = .57, Emotional Control α = .86, Self-Monitor α = .75, Initiate α = . 78, Working Memory α = .77, Plan/Organize α = .76, Task Monitor α = .68, Organization of Materials α = .78.

Table 1
Description of the constructs measured in the BRIEF-A and CogBat.

BRIEF-A	
Behavioral Regulation Index	
Inhibit	Inhibitory control and ability to interrupt ongoing actions
Shift	Ability to change attentional focus adaptively
Self-Monitor	Extent of awareness of one’s behavior and its effects
Emotional Control	Ability to regulate one’s emotions
Metacognition Index	
Initiate	Ability to start planned behavior
Working Memory	Ability to store currently necessary information and to sustain attention
Plan/ Organize	Ability to decide on the appropriate steps to achieve a goal
Task Monitor	Ability to track one’s problem-solving
Organization of Materials	Ability to structure everyday objects
CogBat	
TMT *A*	Processing Speed
B	Cognitive flexibility
B/A	Cognitive flexibility without influence of processing speed
INHIB	Ability to suppress unwanted reactions
TOL-F	Ability to cognitively model solutions for a problem and assess the consequences of an action before it is carried out.

Note. BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; CogBat, neuropsychological test battery for cognitive functions. TMT-A, Trail-making Test Part A; TMT—B, Trail-making Test Part B; TMT-B/A, quotient of Trail-making Test Part A and Part B; INHIB, go/no-go response inhibition task; TOL-F, Tower of London – Freiburg version.

2.3.4. Performance-based assessment of executive functions

We assessed executive functions with three performance-based tests from the CogBat test battery [16] of the Vienna Test System [33]: the Trail Making Test Langensteinbach Version (TMT-L) Parts A and B, the INHIB go/no-go test, and the Tower of London test - Freiburg Version (TOL-F). In the TMT-L, the numbers 1 to 13 and letters A to L are presented on a template on a computer screen in a random pattern, and testees are asked to connect the numbers or numbers and letters in a particular order by clicking on them with the mouse. In Part A, testees connect the numbers as quickly as possible in the correct order; the time needed is assumed to be related to the individual’s information processing speed. In Part B, testees alternately connect the numbers and letters in ascending or alphabetical order; this is assumed to reflect cognitive flexibility, i.e., the ability to switch between different reference systems. Comparing performance in Parts A and B (as a quotient) allows to control for differences between testees that are attributed only to processing speed but are not directly related to executive functions. Unless noted otherwise this quotient was used to assess cognitive flexibility in this study.

We used the INHIB go/no-go paradigm to assess response inhibition, i.e., the ability to suppress irrelevant reactions. In this test, testees are asked to respond to a specific stimulus (e.g., an “x”) as quickly as possible by pressing a key and to not respond to a different stimulus (e.g., a “+”), i.e., to suppress their impulse to respond. The variable commission errors indicates the frequency of unsuccessful inhibitions after no-go stimuli, i.e., the effectiveness of the response inhibition process.

The TOL-F measures planning ability, which can be understood as a form of problem solving and refers to the mental simulation and evaluation of courses of action and the resulting consequences. In a TOL-F task, the testee is asked to move a number of different colored ball-like symbols on three rods to a target arrangement in as few moves as possible, whereby only one ball may be moved at a time. The respondent is instructed to think about the solution first and to move the balls only after they have finished thinking about the task. Planning ability is assessed as the number of tasks solved in the given minimum required moves (see Table 1 for a summary of scales and constructs).

The normative sample of the CogBat included only individuals who had no history of a serious neurological or psychiatric disorder and were not currently taking any medications that affect the central nervous system. The sample comprised 184 (43.9%) men and 235 (56.1%) women aged 16 to 80 years ($M \pm SD$, 42.17 ± 15.75 years). The reliability of the main variables was calculated with various measures: Cronbach's alpha was determined for the TMT and TOL (TMT Part A = .91, Part B = .79, quotient B/A = .68; TOL = .70), whereas the reliability of INHIB was estimated as split-half reliability and was .83.

2.4. Data analysis

Statistical analyses were performed using IBM SPSS Statistics Version 28 and R [34]. Assessing executive functions with standardized tests, i.e., the BRIEF-A and parts of the CogBat test battery, allowed us to compare the performance of forensic psychiatric patients with that of the normative sample. Thus, deviations in cognitive capabilities could be determined for rating- and performance-based instruments with one-sample two-sided *t*-tests. This involved converting the raw scores obtained from the patients sample into T-values and comparing mean T-values with the T-norm value of 50. In the CogBat-test higher scores denote greater abilities, whereas in the BRIEF-A test higher scores denote greater deficits. Correlations between measures of executive functions were analyzed with Pearson's correlation tests. Finally, generalized linear models were calculated to determine contribution of specific executive functions in predicting propensity towards aggression and committed acts of violence. Propensity towards aggression was operationalized as the scores in the two subtests reactive aggression and appetitive aggression of the AFAS [German Version, [29]]. As there was a significant difference between the two subtest scores (t [53] = 6.33, p < .001, Cohen's d = 0.86), the subtests were analyzed separately. Type of offense was operationalized as a binary variable coding offenses that involved acts of violence versus offenses that did not involve violent acts. For an overview of reported crimes and classifications as violent or non-violent see Table 2.

3. Results

Table 2 shows the sociodemographic, clinical, and forensic psychiatric characteristics of the participants. The primary diagnosis of all participants was SUD. Details concerning reported secondary diagnoses, education, and committed crimes follow the information provided by the participants.

3.1. Differences in executive function between the clinical and the normative sample

3.1.1. BRIEF-A

In the BRIEF-A, scores were significantly higher in the patients than in the normative sample for the GEC and BRI, but no significant difference was found for the MI. As to the scales in the BRI, a one-sided *t*-test showed that scores were significantly higher in the patients than in the normative sample in the Inhibit scale, Shift scale, and Self-Monitor Scale, but were not significantly different in the Emotional Control Scale. In the MI, scores for four of the five scales were higher or significantly higher in the patients than in the normative sample. In contrast, the Organization of Materials scale score was significantly lower in the clinical sample implying better performance of the clinical sample in this dimension in comparison to the norm (see Table 3).

Table 2
Descriptive statistics of sociodemographic, clinical, and forensic data as provided by the participants by self-report.

	n (percentage)
Secondary Diagnosis	
Depression	5 (8%)
Personality Disorder	17 (27%)
Phobic and other anxiety disorders, obsessive-compulsive disorder	0 (0%)
Schizophrenia	1 (2%)
Others	4 (6%)
Crimes	
Arson^{++}	1 (2%)
Assault and battery^{+}	12 (19%)
Fraud^{++}	5 (8%)
Manslaughter/murder^{+}	2 (3%)
Robbery^{+}	7 (11%)
Sexual offense^{+}	1 (2%)
Theft^{++}	12 (19%)
Traffic offense^{++}	4 (6%)
Violation of the narcotics act^{++}	33 (52%)
Others	4 (6%)
Education [cf. 49]	
No formal Diploma	12 (19%)
Primary Education	29 (45%)
Lower Secondary Education	17 (27%)
Upper Secondary Education	6 (9%)

Note. $^{+}$ classified as violent, $^{++}$ classified as non-violent.

3.1.2. CogBat

Data for the TMT Part A were missing for five patients. The results of the remaining 59 patients did not differ significantly from the normative sample. In the TMT Part B, we found no significant difference between the patient group and the normative sample, and in the quotient of Parts A and B, the patients were also not different from the normative sample. In the INHIB go/no-go task, the clinical sample showed significantly more commission errors than the normative sample. In the TOL-F, we found no significant difference between the clinical and normative samples (see Table 4).

Table 3
One-sample two-sided t-tests comparing results of the clinical sample in the BRIEF-A with the norm samples.

BRIEF-A	*M*	*df*	*t*	Significance	*Cohen's d*
GEC	53.47	56	3.24	$p = .002$**	0.43
BRI	55.49	56	4.88	$p < .001$***	0.65
Inhibit	58.58	56	7.61	$p < .001$***	1.01
Shift	55.46	56	4.79	$p < .001$***	0.63
Self-Monitor	54.67	56	3.14	$p = .003$**	0.42
Emotional Control	51.11	56	0.94	$p = .352$	0.12
MI	51.61	56	1.41	$p = .166$	0.19
Initiate	52.32	56	1.71	$p = .092$	0.23
Working Memory	52.11	56	1.66	$p = .102$	0.22
Plan/Organize	51.91	56	1.61	$p = .113$	0.21
Task Monitor	55.40	56	4.76	$p < .001$***	0.63
Organization of Materials	46.75	56	−2.91	$p = .005$**	0.39

Note. * $p < .05$, ** $p < .01$, *** $p < .001$; BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; GEC, Global Executive Composite; BRI, Behavioral Regulation Index; MI, Metacognition Index.

Table 4
One-sample two-sided t-tests comparing results of the clinical sample in the CogBat with the norm samples.

CogBat		*M*	*df*	*t*	Significance	*Cohen's d*
TMT	*A*	48.39	58	1.43	$p = .158$	0.19
	B	49.23	63	0.61	$p = .546$	0.08
	B/A	50.90	58	0.66	$p = .513$	0.09
INHIB		44.81	63	3.56	$p < .001$ ***	0.44
TOL-F		48.95	63	1.01	$p = .318$	0.13

Note. * p < .05, ** p < .01, *** p < .001; CogBat, neuropsychological test battery for cognitive functions; TMT-A, Trail-making Test Part A; TMT—B, Trail-making Test Part B; TMT-B/A, quotient of Trail-making Test Part B and Part A; INHIB, go/no-go response inhibition task; TOL-F, Tower of London – Freiburg version.

3.2. Correlations between performance-based and rating measures of executive functions

3.2.1. Superordinate scales BRIEF-A and CogBat

Because the GEC is the sum of the BRI and MI, we excluded it from further analyses of the BRIEF-A results. None of the performance-based assessments (i.e., quotient of Parts A and B of the TMT, TOL-F, and INHIB) correlated significantly with the BRI or the MI of the BRIEF-A (see Table 5).

3.2.2. Basic scales of BRIEF with corresponding CogBat scales

Three basic scales in the BRIEF-A (Shift, Inhibit, and Plan/Organize) map conceptually to the three CogBat tests (see Table 5). We found no significant correlations between these three BRIEF-A scales and the respective CogBat test.

3.3. Subjective and objective measures of executive functions as predictors for subjective ratings of aggression and objective delinquency

To estimate whether subjective and objective measures of executive functions predict the tendency to violent behavior or actually committed acts of violence, we calculated generalized linear models. In the analysis of the subtest *reactive aggression*, we included the three CogBat subtests as predictors in the model. The results showed a significant main effect of performance in the quotient of the TMT Parts A and B but no effect for the number of commission errors or planning capacity (see Table 6a). In the analysis of rating measures of executive functions, we also used a generalized linear model and included the BRIEF-A subtests Inhibit, Shift, and Plan/Organize (i.e., the three subtests that corresponded to the subtests of the CogBat) as predictors. A higher score on the BRIEF-A Inhibit subtest significantly increased the attraction to violence in the sense of reactive aggression, but the Shift and Plan/Organize subtests showed no significant effects (see Table 6a).

Table 5
Correlations between self-rating (Behavior Rating Inventory of Executive Function – Adult Version) and performance-based measures (neuropsychological test battery for cognitive functions, CogBat) of executive functions.

	Performance-based measures (CogBat)		
	TMT - Cognitive Flexibility	INHIB - Inhibition	TOL - Planning capacity
Self-rating (BRIEF-A)			
Behavioral Regulation Index	.03, $p = .80$	−.22, $p = .11$	−.17, $p = .22$
Metacogniton Index	−.06, $p = .66$	−.18, $p = .19$	−.12, $p = .37$
Shift scale	.23, $p = .09$		
Inhibit scale		.07, $p = .61$	
Plan/Organize scale			−.13, $p = .34$

Note. $N = 57$; * p < .05, ** p < .01, *** p < .001; BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; CogBat, neuropsychological test battery for cognitive functions; INHIB, go/no-go response inhibition task; TMT, quotient of Trail-making Test Part B and Part A; TOL-F, Tower of London – Freiburg version.

Table 6a
Results of generalized linear models estimating reactive aggression.

	b	SE(*b*)	Significance
Model Performance-based measures (CogBat)			
TMT - Cognitive Flexibility	0.53	0.15	$p < .001$***
INHIB - Inhibition	0.12	0.14	$p = .413$
TOL - Planning capacity	−0.06	0.20	$p = .782$
Model Self-assessment (BRIEF-A)			
Inhibit scale	0.49	0.16	$p = .003$**
Shift scale	0.25	0.17	$p = .151$
Plan/Organize scale	0.02	0.17	$p = .906$

Note. * p < .05, ** p < .01, *** p < .001; BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; CogBat, neuropsychological test battery for cognitive functions; INHIB, go/no-go response inhibition task; TMT, quotient of Trail-making Test Part B and Part A; TOL-F, Tower of London – Freiburg version.

The analysis of the *appetitive aggression* subtest as a function of performance-based measures revealed a significant main effect of performance in the quotient of TMT Parts A and B, but the number of commission errors (INHIB) and planning capacity (TOL-F) had no significant effect. Appetitive aggression was also significantly predicted by the BRIEF-A Inhibit score, whereas the Shift and Plan/Organize scores showed no significant effects (see Table 6b).

The analysis of type of offense, violent versus non-violent offenses (see Table 2), as a function of performance-based measures revealed no significant main effect of performance on any of the studied scales, i.e., quotient of TMT Parts A and B, number of commission errors (INHIB) and planning capacity (TOL-F) (see Table 7). The analysis of the effect of rating measures on the type of offense showed a significant effect of self-rated deficits in the BRIEF-A Inhibit scale, whereas the Shift and Plan/Organize scales showed no significant effects (see Table 7).

We found highly significant correlations between the type of offense and the self-rated scores for reactive aggression (r [54] $= .45$, $p < .001$) and appetitive aggression (r [53] $= .47$, $p < .001$), showing a positive relationship between attraction to violence and actually committed offenses of physical violence.

4. Discussion

Impairments in executive functions are considered to be a relevant deficit in patients with SUD [1] and a relevant factor in the analysis of aggression [7]. In our study we analyzed data of patients with SUD who were convicted of crimes committed in the context of their disorder. We assessed executive functions in this patients group with self-report and

Table 6b
Results of generalized linear models estimating appetitive aggression.

	b	SE(*b*)	Significance
Model Performance-based measures (CogBat)			
TMT - Cognitive Flexibility	0.39	0.13	$p = .003$**
INHIB - Inhibition	0.09	0.12	$p = .467$
TOL - Planning capacity	−0.05	0.16	$p = .770$
Model Self-assessment (BRIEF-A)			
Inhibit scale	0.59	0.15	$p < .001$***
Shift scale	0.09	0.15	$p = .553$
Plan/Organize scale	0.01	0.15	$p = .\ 947$

Note. * p < .05, ** p < .01, *** p < .001; BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; CogBat, neuropsychological test battery for cognitive functions; INHIB, go/no-go response inhibition task; TMT, quotient of Trail-making Test Part B and Part A; TOL-F, Tower of London – Freiburg version.

Table 7
Results of generalized linear models estimating violent offenses (0 = no violent offense, 1 = violent offense).

	b	SE(b)	Significance
Model Performance-based measures (CogBat)			
TMT - Cognitive Flexibility	0.04	0.03	$p = .180$
INHIB - Inhibition	0.01	0.03	$p = .770$
TOL - Planning capacity	−0.08	0.04	$p = .091$
Model Self-assessment (BRIEF-A)			
Inhibit scale	0.10	0.04	$p < .016$*
Shift scale	−0.02	0.04	$p = .549$
Plan/Organize scale	0.03	0.04	$p = .419$

Note. * p < .05, ** p < .01, *** p < .001; BRIEF-A, Behavior Rating Inventory of Executive Function – Adult Version; CogBat, neuropsychological test battery for cognitive functions; INHIB, go/no-go response inhibition task; TMT, quotient of Trail-making Test Part B and Part A; TOL-F, Tower of London – Freiburg version.

performance-based measures. Additionally, questionnaires on propensity towards aggression and on type of crime, violent versus non-violent, were applied. This multiperspective approach allowed us to evaluate correlations between these measures and to test the predictive functions of the studied constructs. It also served as the basis for evaluating the validity of self-assessment in forensic psychiatric patients.

4.1. Lower executive functions in the clinical sample

Our data showed significant differences between the clinical and the normative sample in one of the three performance-based measures, response inhibition, where the clinical sample showed significantly worse inhibition performance. There were no differences in the scales cognitive flexibility and planning capability. The self-ratings in the BRIEF-A differed in the majority of the scales significantly or close to significantly from the normative sample. These ratings might reflect actual impairments that patients experience in their everyday activities. Items of the BRIEF-A are designed for high ecological validity and target deficits that occur in everyday situations. To answer the questions, testees consider their actual experiences to check whether the item describes them adequately. The participants in our study rated themselves as impaired in the Inhibit, Shift, Self-Monitor, and Task Monitor scales, which operationalize executive functions that may be an integral part in modelling the cognitive preconditions of SUD. Interestingly, patients assessed themselves as less capable in all dimensions except the Organization of Materials scale, where patients rated their capability as significantly higher than the normative sample. Further studies are necessary to confirm this effect, but a preliminary hypothesis could be that SUD necessitates skills in this dimension. Most patients with SUD have lived with their disorder for years, so they might be well aware of what is required to sustain an addiction. Among these requirements might be a high capability in efficiently organizing their belongings keeping them “readily available for their use” [15].

4.2. Subjective and objective measures

We compared self-assessments and objective measures of executive functions because of the relevance of these functions in forensic psychiatric patients [1,7]. Although previous studies found some substantial differences between subjective and objective measures [e.g. 8,12,13], both types of assessment are assumed to assess the same cognitive domain. Toplak et al. [12] argue that the two approaches tap into different levels of analysis. However, our study focused less on the question whether these two approaches can be used to validate each other but more on the question whether they can complement each other and provide information that can only be gained when self-rating and performance-based measures are used together. We found no direct correlations between the self-rating and performance-based measures of executive functions included in our study. Generally, being asked to assess one’s mental capabilities, as in the BRIEF-A test, requires complex introspective processes, and subjects must form a concept of the specific capability. Individual variations certainly exist in such subjective concepts because of differences in individuals’ respective experiences. Furthermore, subjects must remember representative episodes in which the relevant mental processes were active. Based on these memories subjects need to find a quantifiable description of the effectiveness of the specific process in action and finally to compare their performance with the performance of other individuals. In light of this complex process, it is remarkable that self-assessments can predict performance at all, although they never do so completely or for every mental process. Our data showed no correlation between self-assessment and performance measures. This finding corroborates those of similar studies in other contexts [8,12,13].

Rothlind et al. [10] demonstrated that their subjects’ assessments of their own performance correlated with their performance in standardized neuropsychological tests, but the correlations were modest. They report that correlations were higher if the tasks had an immanent feedback structure and were thus informative about success or failure. There was no such direct relationship between the measures used in our study. The naturalistic approach of the BRIEF-A thus comes with a cost: The capabilities tested are linked to the situational conditions that represent the ecological background of each item. Consequently, although instruments such as the BRIEF-A validly inform about limitations due to deficits in executive functions, their ecological validity limits their function as a measure of pure cognitive processes.

4.3. Aggression and violent criminality

Scales from both types of tests were predictive of a propensity to aggressive behavior as well as an actual willingness to commit crimes of physical violence. The extent of self-rated inhibitory deficits correlated with a higher tendency to aggression. Patients who rated their inhibitory capacity as low were also significantly more likely to have been convicted of a crime involving violent behavior. The interpretation of the relationship between aggression and inhibitory capacity appears to be straightforward: Impulses to act on anger are more likely to be followed if inhibitory barriers are lower [35]. The executive function of response inhibition resembles the concept of disinhibition – a domain in the personality pathology in the quantitative classification movement [36]. Disinhibited externalizing is integrated in the Hierarchical Taxonomy of Psychopathology Model (HiTOP) as one of six spectra and it is associated with substance abuse and antisocial behavior. The HiTOP model describes psychopathology by quantifiable dimensions of clinical and personality disorders. The difference between disinhibition as an executive function and as a construct in personality pathology or quantitative classification is a matter of abstraction. The concept of executive function is aimed at describing a concrete mental process. Disinhibition as a spectrum in the HiTOP system or in personality pathology describes a trait as a result of many mental processes having taken place over longer periods of time. There is a structural similarity between these concepts but this similarity is neither necessary nor sufficient. Thus, a person characterized by the disinhibition trait may act impulsively but this may or may not be due to deficient executive functions. Young et al. [37] studied the relationship of different executive functions with disinhibition in adolescents and found that response inhibition was most closely related to it but that task-set shifting or working memory updating also correlated with disinhibition. Further research is necessary to study the relationship between (dis-)inhibition as an executive function and as a trait or scale; our results, however, show that the association found for disinhibition as a trait or scale with substance abuse and antisocial behavior can be demonstrated at the level of basic mental processes.

Aggressive tendencies also correlated with the performance-based measure of cognitive flexibility, with higher flexibility predicting a higher propensity towards aggression. The positive correlation between cognitive flexibility and aggression is surprising. One could assume that cognitive flexibility facilitates conflict-free behavior because the capacity to think flexibly could help an individual to find non-violent solutions to problems. In a review by Burgess [38] higher cognitive flexibility as measured with the TMT was actually found to be predictive of less aggression. The result in our study could be related to findings that have been summarized in attention-related theories of psychopathy. Response Modulation Hypothesis [39], Attention Bottleneck Model of Psychopathy [40], and Impaired Integration Theory [41] are based on experiments demonstrating differences in selective attention between psychopathic individuals and controls. For example, psychopathic offenders have been shown to suffer less interference in specific stroop tasks [42] or in an attentional blink task [43]. Unfortunately, assessment of psychopathy was not available for our patient sample. So, this possibility of explanation can only be preliminary. Further research is necessary to confirm if there is a causal connection between performance in tasks affording high cognitive flexibility, propensity to aggression, and psychopathy.

4.4. Bias in self-assessment

In our sample, participants rated themselves as less capable in more dimensions than they actually proved to be as was demonstrated in performance-based measures. Only inhibitory functions were impaired in both performance-based and self-rated tests. In the other performance-based measures patients did not differ from the normative sample. However, the self-ratings in the clinical sample were consistently worse than in the normative sample – with the exception of the Organization of Materials scale as discussed above. Two possible hypotheses could explain this finding. Firstly, SUD could impair subjective conscious experience of cognitive processes. Moeller and Goldstein [44] found evidence that self-awareness is impaired in patients with a SUD, which may mean that SUD specifically leads to deficits in introspection into mental processes. Moeller and colleagues [45] assumed that specific structural differences in the brain are causal for metacognitive impairments in the case of cocaine users. There are also a number of studies showing that individuals with SUD display problems with insight into their disorder and its consequences [46–48]. Thus, our results could reflect this potential metacognitive impairment by comparison of subjective and objective methods of assessment of the cognitive functions possibly affected by SUD. The second hypothesis refers to the measurement instruments themselves: The lack of correlation between self-rated and performance-based assessments may simply be due to differences in how comprehensive or specific the instruments are. While, subjective measures target impairments in everyday functioning that individuals notice consciously, objective measures target the functioning of cognitive routines that often run automatically and without conscious monitoring. Although these cognitive routines may be part of the more complex everyday functioning, they may not be essentially necessary for their success [20]. Thus, some deficits that are registered in one instrument simply cannot register in the other.

4.5. Limitations

Intelligence is potentially confounded with executive functions. However, as this study was part of a larger project [30], assessing IQ would have overtaxed participants.

We did not check for influence of the secondary diagnosis. The patients in the sample were hospitalized primarily because of their SUD and we assumed that it was this diagnosis that mainly influenced cognitive functions.

A further limitation is sample size. The size of the sample reported in this study was due to limited availability of patients in treatment in forensic-psychiatric hospitals. However, larger samples would provide sufficient power for analyses including more variables like IQ, age, or education.

4.6. Conclusions

In sum, our study demonstrates that by using tests from both traditions in parallel, information can be gathered that would not be attainable elsewise. Parallel use of performance-based and self-rated assessments can inform the practitioner whether patients are aware of central deficits that can be associated with increased risks of aggression and violent behavior. Therefore, only a careful inspection of information of both sources can help establish a comprehensive impression of the individual patient and a thorough assessment of possible risks.

Ethics statement

The study was approved by the local ethics committee (Ulm University, Ulm, Germany, No. 497/18).

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CRediT authorship contribution statement

Sebastian Pichlmeier: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Judith Streb:** Formal analysis, Conceptualization. **Franziska Anna Rösel:** Investigation, Data curation. **Hannah Dobler:** Investigation, Data curation. **Manuela Dudeck:** Conceptualization. **Michael Fritz:** Methodology.

Declaration of competing interest

The authors declare no conflicts of interest.

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Data availability

The raw data supporting the conclusions of this manuscript will be made available by the authors, without undue reservation, to any qualified researcher. Requests to access the datasets should be directed to judith.streb@uni-ulm.de.

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